

Tutorial 10: Applied Statistical Epidemiology with R

BIOL90042

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In this tutorial, learners should be able to:

1. Clean and prepare epidemiological data
2. Estimate infection prevalence (the proportion of individuals in a population who are infected at a given time)
3. Test for associations between categorical variables (chi-squared test)
4. Choose between parametric (t-test, ANOVA) and non-parametric (Wilcoxon, Kruskal-Wallis) tests for a continuous outcome
5. Fit a linear model and check its assumptions

Background

Epidemiology

Epidemiology is the study of diseases in populations. It is an extension or application of statistics and investigates the patterns of disease (e.g., infection or non-communicable) and other illnesses to help protect the health and safety of our communities.

Epidemiologists look at patterns, trends and data related to a disease or illness and can establish risk assessments or predictions of what to expect next. Epidemiologists use evidence and data to make informed decisions in public health.

Malaria and mass drug administration

Plasmodium falciparum malaria remains a leading cause of childhood mortality in sub-Saharan Africa. **Artemisinin-based combination therapies (ACTs)** are the first-line treatment for malaria. In our hypothetical scenario, substantial funding has recently been directed toward administering ACTs as a **blanket strategy** rather than reactive treatment of symptomatic cases — an approach called **mass drug administration (MDA)**, where everyone in a population is treated regardless of infection status.

Other control tools include **insecticide-treated bed nets (ITNs)**.

Study scenario

Researchers surveyed a village **twice**: **survey_1**, conducted just before an MDA program was rolled out, and **survey_2**, conducted in the same village about 12 months later, while the program was running. The same participants were seen in both surveys. At each visit, researchers recorded demographic information, intervention use, current infection status (by PCR) and parasite density.

Research question: *Does MDA with ACT influence the burden of malaria?*

We break this down into:

- Does it influence the **prevalence** of infection?
- In **whom** (which age groups)?
- Does it influence the **severity** of infection (parasite density)?

Setup

`malaria_participants.csv` contains one row per participant **per survey** with the following information:

- **id**: identifier per person
- **survey**: survey that the information was obtained: "survey_1" (before MDA) or "survey_2" (during MDA)
- **age**: participant-identified age
- **sex**: participant-identified sex
- **pcr_positive**: 18S rRNA PCR result: 1 = currently infected, 0 = not infected
- **parasite_density**: parasites per microlitre of blood (0 if PCR negative)
- **act_use**: artemisinin-based combination therapy provided to participant: 1 = yes, 0 = no
- **bednet_use**: slept under a bed net the previous two weeks: 1 = yes, 0 = no

You can analyse the data by either looking at one survey at a time (`filter(survey == "survey_2")`) or explicitly comparing the two surveys.

```
library(tidyverse)

participants <- read_csv(
  "https://raw.githubusercontent.com/WEHI-Education/BIOL90042_R_Course/refs/heads/main/data/malaria_par"
)
```

Part 1: Data cleaning

Clean up the data:

1. Remove rows with NAs using `na.omit()` or `drop_na()`
2. Use `mutate(case_when())` to recode `pcr_positive` into words, where 1 is "yes" and 0 is "no" (call it `pcr_status`)
3. Use `mutate(case_when())` to recode `age` into groups 0-5, 6-10, 11-20, 21+ (call it `age_group`)
4. How many rows remain? How many in each survey?

Part 2: Does MDA influence the prevalence of infection?

What was the prevalence of infection in each survey? *Hint*: Prevalence = number PCR positive / total tested. You will need to use the `group_by()` and `summarise()` functions to execute this calculation.

Is infection status associated with survey? Use a chi-squared test to evaluate the null hypothesis that PCR status and survey are independent. *Hint* Revisit your workshop 6 materials for the correct function/s to use!

Discuss

- Does prevalence appear to change between the two surveys? Is this supported by the statistical test?
- What assumptions does the chi-squared test make? Are they met here?
- Is a before/during comparison on its own enough to attribute the change to MDA? What else could explain it?

Part 3: In whom? Prevalence by age group

Calculate prevalence by age group within each survey - by eye, does the prevalence of infection vary between age groups and over time?

Next, filter to `survey_2` and run a chi-squared test of age group by infection status to test the null hypothesis that PCR status and age group are independent.

Discuss

- What type of variables are we testing here? Why is a chi-squared test appropriate?
- Which age group has the highest prevalence? Does the association hold in both surveys?
- Because MDA is a *blanket* intervention (everyone is treated), we cannot compare treated vs untreated people. What comparison are we making instead, and what are its limits?

Part 4: Does MDA influence the severity of infection?

Severity is measured by `parasite_density` among **infected individuals** (i.e., those who are `pcr_positive == 1`). PCR-negative participants have a density of 0 by definition, so exclude them first and save them into a new variable called `infected`.

Before choosing a test, check the normality assumption **within each group** using a QQ-plot and a Shapiro-Wilk test. *Hint* Revisit your workshop 6 materials for the correct function/s to use!

i Discuss

- Based on the QQ-plot and Shapiro-Wilk test, is a t-test valid here?
- Which test should you use instead?

Now run both the parametric and non-parametric test and compare. To ask whether severity also differs **across age groups** (>2 groups), the parametric option is a one-way ANOVA and the non-parametric option is the Kruskal-Wallis test.

i Discuss

- Does parasite density among the infected differ between surveys? Between age groups?
- How does severity declining with age fit with what you know about acquired immunity to malaria?

Part 5: A linear model

Comparing groups one at a time is limited. A **linear model** lets us test a formal hypothesis while adjusting for other variables. Our hypothesis:

Among infected individuals, log parasite density declines with age and is lower during MDA (survey_2).

We model `log(parasite_density)` because parasite densities span orders of magnitude and are right-skewed; logging makes the relationship closer to linear and the residuals closer to normal.

```
model_density <- lm(log(parasite_density) ~ age + survey, data = infected)
summary(model_density)

# exponentiate coefficients to interpret them as multiplicative effects
exp(coef(model_density))
```

i Discuss

- What does the `age` coefficient mean after exponentiating? (Hint: a value of ~0.96 means density falls by ~4% per year of age.)
- What does the `surveysurvey_2` coefficient (~0.40) tell you about the effect of MDA on severity, at the same age?

Now check the model assumptions (linearity, homoscedasticity, normality of residuals):

Hint Revisit your workshop 6 materials for the correct function/s to use!

```
# residuals vs fitted: want no pattern, constant spread

# QQ-plot of residuals: points should sit along the reference line

# Run the shapiro test for residuals
```

i Discuss

- In the residuals-vs-fitted plot we want no pattern and constant spread; in the QQ-plot we want the points to follow the diagonal line. What do we see in our plots?
- Does the Shapiro-Wilk test on the residuals agree with the QQ-plot? (Hint: check whether the p value is above or below 0.05.)

Answers

Part 1:

```
participants <- participants %>%
  # you can use na.omit() or drop_na()
  na.omit() %>%
  # create your two new columns
  mutate(
    # pcr status is just pcr_positive but re-coded as a character
    pcr_status = case_when(
      pcr_positive == 1 ~ "yes",
      TRUE ~ "no"
    ),
    # age group also uses numeric values and creates a character
    age_group = case_when(
      age <= 5 ~ "0-5",
      age <= 10 ~ "6-10",
      age <= 20 ~ "11-20",
      TRUE ~ "21+"
    )
  )

participants %>% count(survey)
```

```
# A tibble: 2 x 2
  survey      n
  <chr>    <int>
1 survey_1  885
2 survey_2  890
```

25 rows were excluded, all because `pcr_positive` (and therefore `parasite_density`) was missing. That is only ~1.4% of the data, and the missingness is spread across groups rather than concentrated in one, so dropping them is unlikely to bias the results.

Part 2:

```
# calculate the prevalence of infection per survey
participants %>%
  group_by(survey) %>%
  summarise(
    n_tested = n(),
    n_positive = sum(pcr_positive),
    prevalence = n_positive / n_tested
  )
```

```
# A tibble: 2 x 4
  survey  n_tested n_positive prevalence
  <chr>    <int>    <dbl>    <dbl>
1 survey_1    885      275     0.311
2 survey_2    890      150     0.169
```

```
# perform chi-squared test to evaluate the null hypothesis that PCR status and survey are independent.
survey_pcr_table <- table(participants$survey, participants$pcr_status)
chi_sq <- chisq.test(survey_pcr_table)
chi_sq$expected
```

```
      no      yes
survey_1 673.0986 211.9014
survey_2 676.9014 213.0986
```

chi_sq

Pearson's Chi-squared test with Yates' continuity correction

```
data: survey_pcr_table
X-squared = 48.492, df = 1, p-value = 3.317e-12
```

Prevalence fell from ~31% in survey_1 to ~17% in survey_2, and the chi-squared test shows this is highly statistically significant ($p < 0.001$). The expected counts are all large (well above 5), observations are independent, and categories are mutually exclusive, so the test is valid. However, a before/during comparison alone cannot prove MDA caused the decline — a particularly dry year, bed net campaigns, or other changes could also reduce transmission. The before/during design is stronger than a single snapshot but weaker than a randomised trial.

Part 3:

```
participants %>%
  group_by(survey, age_group) %>%
  summarise(prevalence = mean(pcr_positive))

# A tibble: 8 x 3
# Groups:   survey [2]
  survey age_group prevalence
  <chr>   <chr>         <dbl>
1 survey_1 0-5          0.554
2 survey_1 11-20         0.289
3 survey_1 21+          0.186
4 survey_1 6-10          0.457
5 survey_2 0-5          0.279
6 survey_2 11-20         0.170
7 survey_2 21+          0.117
8 survey_2 6-10          0.238

survey_2 <- participants %>% filter(survey == "survey_2")
chisq.test(table(survey_2$age_group, survey_2$pcr_status))
```

Pearson's Chi-squared test

```
data: table(survey_2$age_group, survey_2$pcr_status)
X-squared = 25.145, df = 3, p-value = 1.44e-05
```

We are testing two categorical variables (age group and PCR status), so a chi-squared test is appropriate.

The youngest children (0-5) have the highest prevalence in both surveys (~55% before, ~28% during), and prevalence declines with age — consistent with acquired immunity developing with exposure.

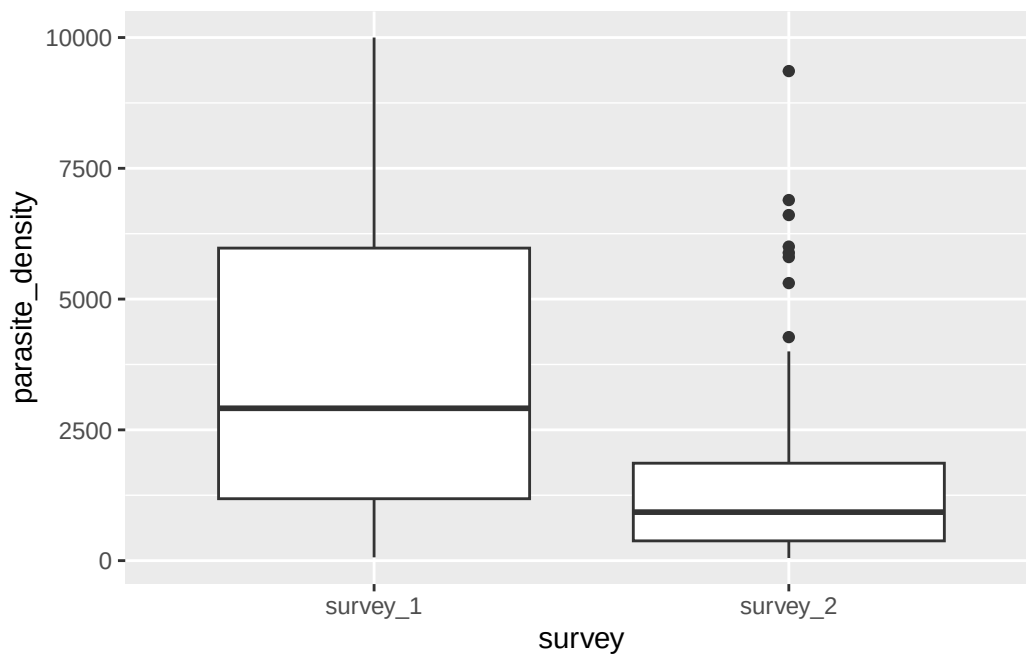
The association between age group and infection is significant in both surveys ($p < 0.001$).

Because MDA treats everyone, there is no untreated comparison group: we can only compare before vs during, within and across age groups, and any village-wide changes over the year remain a possible alternative explanation.

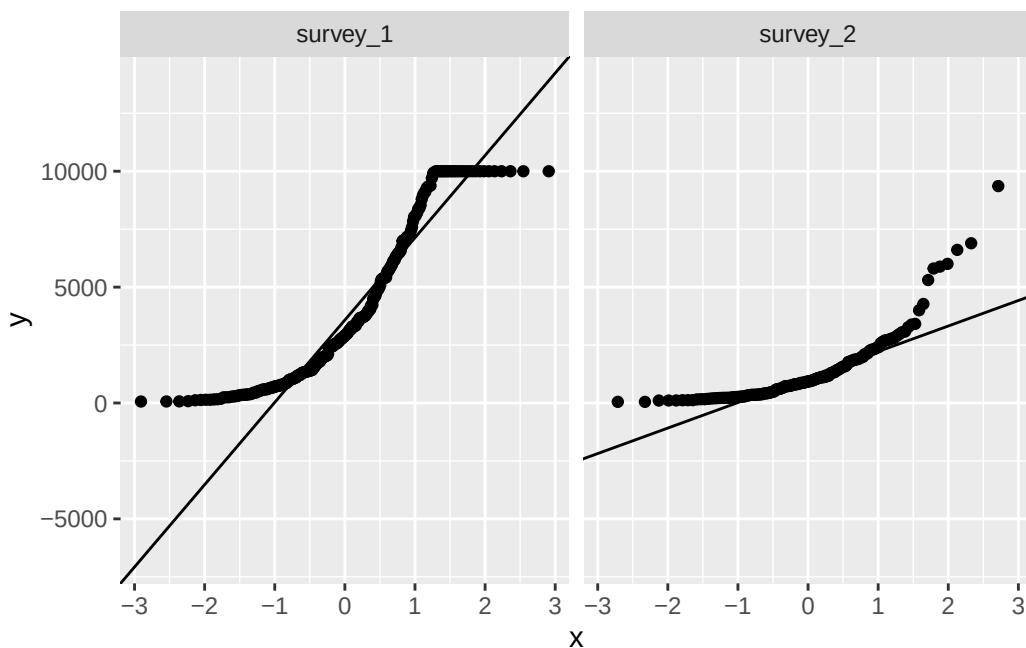
Part 4:

```
infected <- participants %>% filter(pcr_positive == 1)

ggplot(infected, aes(x = survey, y = parasite_density)) +
  geom_boxplot()
```



```
ggplot(infected, aes(sample = parasite_density)) +
  geom_qq() +
  geom_qq_line() +
  facet_wrap(~survey)
```



```
infected %>%
  filter(survey == "survey_1") %>%
  pull(parasite_density) %>%
  shapiro.test()
```

Shapiro-Wilk normality test

```
data: .
W = 0.88002, p-value = 7.018e-14
```

The QQ-plots curve away from the reference line and the Shapiro-Wilk p value is far below 0.05 in both surveys: parasite density is **not** normally distributed, so the Wilcoxon rank-sum test is the appropriate choice.

```
t.test(parasite_density ~ survey, data = infected, var.equal = TRUE)
```

Two Sample t-test

```
data: parasite_density by survey
t = 8.8715, df = 423, p-value < 2.2e-16
alternative hypothesis: true difference in means between group survey_1 and group survey_2 is not equal
95 percent confidence interval:
 1900.332 2982.089
sample estimates:
mean in group survey_1 mean in group survey_2
      3836.051           1394.840
```

```
wilcox.test(parasite_density ~ survey, data = infected)
```

Wilcoxon rank sum test with continuity correction

```
data: parasite_density by survey
W = 30983, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal to 0
```

```
aov(parasite_density ~ age_group, data = infected) %>% summary()
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
age_group	3	1.024e+09	341194470	53.93	<2e-16 ***
Residuals	421	2.664e+09	6326886		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
kruskal.test(parasite_density ~ age_group, data = infected)
```

Kruskal-Wallis rank sum test

```
data: parasite_density by age_group
Kruskal-Wallis chi-squared = 183.74, df = 3, p-value < 2.2e-16
```

Both tests agree: parasite density among the infected is significantly **lower** in survey_2 ($p < 0.001$). Density also differs significantly across age groups (ANOVA and Kruskal-Wallis both $p < 0.001$), declining with age — consistent with acquired immunity reducing parasite growth in older, more-exposed individuals. So MDA influenced both the **prevalence** and the **severity** of infection.

Part 5:

- The **age** coefficient exponentiates to ~0.96: at the same survey, expected parasite density falls by ~4% per additional year of age.
- The **surveysurvey_2** coefficient exponentiates to ~0.40: at the same age, expected density during MDA is ~40% of what it was before — a ~60% reduction in severity among those still infected.

```
# residuals vs fitted: want no pattern, constant spread
ggplot(
  tibble(fitted = fitted(model_density), resid = residuals(model_density)),
  aes(x = fitted, y = resid)
) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed")

# QQ-plot of residuals: points should sit along the reference line
ggplot(tibble(resid = residuals(model_density)), aes(sample = resid)) +
  geom_qq() +
  geom_qq_line()
```

```
# Run the shapiro test for residuals  
shapiro.test(residuals(model_density))
```

- Both look good here, with only slight deviations at the extreme tails of the QQ-plot.
- The Shapiro-Wilk p value is ~ 0.18 , above 0.05, so we fail to reject the null hypothesis that the residuals are normally distributed. This agrees with the QQ-plot: both checks support the normality assumption, so the p values and confidence intervals from the model are trustworthy.